Registration & Validation Engine: Complete Guide

Architectural Breakdown of Age, Gender, Profile Readiness, and Pre-Submit Validation in **SportsDashboard.tsx** and **SportsRegister.tsx**

Target Modules: SportsDashboard.tsx, SportsRegister.tsx, SportsMultiRegister.tsx | Status: Production Ready **Date: October 2026**

Executive Summary: The 3-Tier Validation Pipeline
The Mana Community sports registration engine provides multi-layered validation guarantees to completely prevent ineligible participation, age fraud, gender mismatches, profile bypasses, and double registrations across single and multi-sport tournaments:
• **Tier 1 - Real-Time Dashboard Eligibility Matrix:** Automatically evaluates participant age and gender across every event card before any click occurs. Cards render reactive states (Registered, Partial, Blocked, Complete Profile, or Register Now) and dynamically hide ineligible categories when smart filtering is active.
• **Tier 2 - Registration Modal Identity & Sibling Guard:** Enforces full profile completeness (Name, Gender, verified DOB, Flat/Villa), displays real-time warning banners, and disables ineligible sibling category tiles with click interception.
• **Tier 3 - Pre-Submit Checkout & Verification Gatekeeper:** Mandatory DOB proof document validation for upper-age brackets (≤ 5MB), mandatory doubles partner allocation, duplicate registration lock, optional email OTP, reCAPTCHA, and Cashfree payment confirmation.

1. Core Engine: checkCategoryEligibility() Deep Dive

Defined in SportsRegister.tsx (L128) and mirrored in SportsMultiRegister.tsx, this pure evaluation function processes the candidate event against participant age and gender with mathematical precision.

Evaluation Stage	Logic & Extraction Strategy	Validation Rules & Failure Conditions	Resulting UI Badge
1. Age Bounds Extraction	Reads minAge and maxAge with safe numeric coercion (Number(...)). If null, parses title with regex: • Under-X: /(?:under|u-?|below)\s*(\d+)/i • X+: /(\d+)\s*(?:\+|plus|above)/i • Range: /(\d+)\s*[—to]\s*(\d+)/i • Kids: Max 16y | Seniors: Min 45y	• **Upper Age:** If participantAge > maxAge → Flags isOverAge = true. • **Lower Age:** If participantAge < minAge → Flags isUnderAge = true. • **General Sports:** If age < 4 and not baby event → Flags isUnderAge = true. • **Zero/Missing Age Lock:** If age ≤ 0 and category requires minAge > 0 → Flags isUnderAge = true (DOB required).	**Over Age Limit or Under Age Limit (eligible: false)**
2. Gender Requirement	Reads explicit DB gender (MALE, FEMALE, MIXED, ALL). If null, scans title keywords: • **Female Checked First:** womens?|woman|girls?|ladies • **Male with Negative Lookbehind:** (?<![a-z])(mens?|man\b|males?|boys?) (prevents "men" matching inside "women").	• Normalizes profile gender via normalizeGender(). • If category is MALE and profile is Female → Mismatch. • If category is FEMALE and profile is Male → Mismatch. • MIXED and ALL accept any participant gender.	**Gender Ineligible (isGenderMismatch = true)**
3. Sibling Array Hierarchy	Events frequently contain nested category brackets in e.categories.	• Direct parent restrictions are **ALWAYS** evaluated first. • Subcategory arrays are only inspected if the parent item has no restrictions. • Completely prevents child brackets with no age bounds from overriding an explicit 19+ parent constraint.	**Strict Parent Guard (No bypass possible)**

2. Sports Dashboard: Real-Time Reactive Layer (SportsDashboard.tsx)

The dashboard automatically binds the selected participant to all tournament events. Whenever the user switches participants, the entire screen recalculates immediately without a page refresh.

Participant State / Card Condition	Button / Badge Rendered	Visual Style & CSS Classes	Interactive Behavior & Action
Fully Registered User already registered for all formats of this sport.	**✓ Registered**	`bg-emerald-50 text-emerald-700 border-emerald-200`	Non-clickable confirmed status badge. Prevents duplicate booking.
Partially Registered Registered for Singles, Doubles format remaining.	**✓ Singles + + Doubles**	`bg-indigo-50 text-indigo-700 border-indigo-200`	Clicking opens registration modal pre-targeted to the unregistered Doubles format.
Ineligible Participant Fails min/max age or gender criteria.	**Under Age Limit or Over Age Limit or Gender Ineligible**	`bg-rose-50 text-rose-700 border-rose-200 cursor-not-allowed`	**Replaces "Register Now" completely.** Card dims; hover tooltip displays exact restriction reason.
Profile Incomplete Candidate missing DOB, Age, or Gender.	**Complete Profile →**	`bg-slate-100 text-slate-500 border-slate-200 hover:bg-slate-200`	Opens profile completion modal to enter missing DOB and Gender before proceeding.
Eligible & Ready Participant satisfies age, gender, and profile.	**Register Now ■**	`bg-gradient-to-r from-indigo-600 to-violet-600 text-white`	Opens registration modal pre-filled with verified participant identity.

3. Registration Modal: 7-Stage Pre-Submit Pipeline (SportsRegister.tsx)

When the registration modal opens, it enforces a sequential validation pipeline before allowing any record to be saved or sent for payment.

Pipeline Stage	Component / Code Reference	Exact Validation Mechanism	Enforcement Action
Stage 1: Identity & Profile Gate	`SportsRegister.tsx Lines 1916–1940`	Verifies Full Name, Gender, Flat/Villa, and Date of Birth. Even if age is known, an exact calendar DOB is mandatory.	Halts submission; triggers toast; opens openUpdateDetailsModal.
Stage 2: Sibling Category Guard	`SportsRegister.tsx Lines 3370–3458`	Evaluates sibling brackets in the tournament. Ineligible categories are dimmed (opacity 60%, cursor-not-allowed).	Clicking ineligible tile is intercepted & blocked; triggers toast.error().
Stage 3: Upper-Age Proof Gate	`SportsRegister.tsx Lines 1968–1978`	Upper-age brackets (19+, 45+) require uploading Government or School ID with DOB proof (≤ 5MB file size).	Blocks checkout until valid proof file is attached.
Stage 4: Doubles Partner Mandate	`SportsRegister.tsx Lines 2013–2017`	If Doubles or Mixed Doubles format is selected, user must assign: (A) Resident partner, (B) Family partner, or (C) Open Pool.	Blocks submit: *"Doubles Partner is mandatory. Please select partner or Open Pool."*
Stage 5: Duplicate Booking Guard	`SportsRegister.tsx Lines 1911–1915`	Cross-checks existing active registrations for this participant across all tournament formats.	Blocks submit: *"Participant is already registered for this event."*
Stage 6: Security Verification	`SportsRegister.tsx Lines 1990–2005`	Validates 6-digit email OTP (if enabled on tournament) and gathers Google reCAPTCHA v2 checkbox response token.	Blocks submit if email is unverified or captcha is incomplete.
Stage 7: Payment & Commit Gate	`SportsRegister.tsx Lines 2020–2075`	For paid events, routes through Cashfree Gateway order session. Registration is committed only upon verified payment callback.	Locks record creation until payment confirmation.

4. Concrete Real-World Scenarios Walkthrough

The following table illustrates how the validation engine dynamically reacts across different participant profiles for the same tournament:

Candidate Profile	Event: Badminton Men's 19+	Event: Badminton Women's 19+	Event: Carrom Boys Under-16
1. Sandeep Kamarapu **Male, Age 30y**	**✓ ELIGIBLE** Age 30 ≥ 19, Gender Male. **Button: Register Now ■**	**✗ GENDER MISMATCH** Requires Female, profile is Male. **Badge: Gender Ineligible**	**✗ OVER AGE LIMIT** Age 30 > Max 16y. **Badge: Over Age Limit**
2. Sindhuja Chinthala **Female (Spouse), Age 28y**	**✗ GENDER MISMATCH** Requires Male, profile is Female. **Badge: Gender Ineligible**	**✓ ELIGIBLE** Age 28 ≥ 19, Gender Female. **Button: Register Now ■**	**✗ OVER AGE LIMIT** Age 28 > Max 16y. **Badge: Over Age Limit**
3. Child Member **Male (Son), Age 12y**	**✗ UNDER AGE LIMIT** Age 12 < Min 19y. **Badge: Under Age Limit**	**✗ INELIGIBLE** Gender mismatch + Under age. **Badge: Gender Ineligible**	**✓ ELIGIBLE** Age 12 ≤ 16, Gender Male. **Button: Register Now ■**
4. Unverified Profile **DOB / Age: 0 / Missing**	**✗ AGE UNVERIFIED** DOB required for 19+ events. **Badge: Under Age Limit**	**✗ AGE UNVERIFIED** DOB required for 19+ events. **Badge: Under Age Limit**	**✗ INCOMPLETE PROFILE** DOB / Age missing in profile. **Button: Complete Profile →**

5. Verification & Security Guarantees Summary

Key Platform Guarantees Enforced by this Implementation:

- **Zero False Approvals:** Ineligible age or gender will always resolve elig.eligible = false, replacing "Register Now" with a rose-colored disabled badge.
- **No Missing-Age Bypass:** A participant with unspecified or 0 age is prevented from registering for adult (19+) events; DOB proof is strictly enforced.
- **No Subcategory Leaks:** Parent tournament rules supersede nested subcategory arrays.
- **Partner & Format Integrity:** Doubles entries require verified partners or explicit pool opt-in.
- **Payment Enforced:** Paid events cannot confirm registrations without Cashfree payment callback verification.